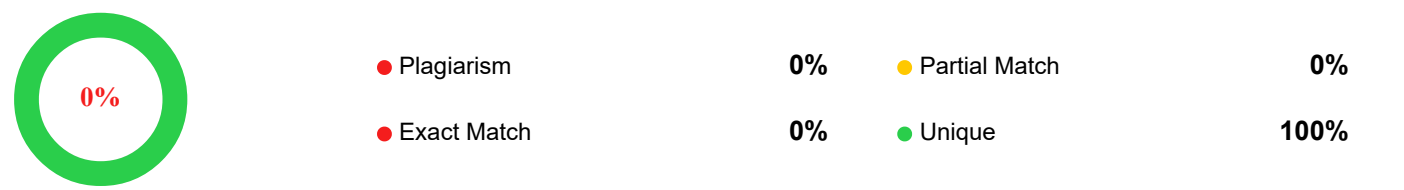


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A Comparative Performance and Stability Analysis of

6T and 7T SRAM Cells for Low-Power Applications

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Abstract—Static random-access memory (SRAM) stability and power efficiency are critical challenges in low-voltage system-on-chip designs. Conventional 6T SRAM cells suffer from degraded read stability and increased sensitivity to process, voltage, and temperature variations at reduced supply voltages. This paper presents a comparative benchmarking analysis of 6T and decoupled-read 7T SRAM cells using static noise margin, access delay, power, and leakage metrics. Circuit-level and Monte Carlo simulations show that 7T SRAM significantly improves read stability and low-voltage robustness compared to 6T SRAM, at the cost of increased area and read delay. The results highlight key trade-offs for selecting SRAM architectures in low-power memory applications.

Keywords—SRAM, 6T SRAM, 7T SRAM, static noise margin, low-voltage operation, benchmarking.

I. INTRODUCTION (HEADING 1)

Static random-access memory (SRAM) is a fundamental

building block in modern digital systems and occupies a significant portion of on-chip area and power consumption in system-on-chip (SoC) designs. SRAM is widely used in cache memories, register files, and embedded memory blocks due to its high speed and compatibility with CMOS processes. As technology scales and power-efficient computing becomes increasingly important, SRAM operation at reduced supply voltages has emerged as a critical design requirement. However, aggressive voltage scaling significantly degrades SRAM stability and reliability, making robust memory design a major challenge in low-power applications.

The conventional 6T SRAM cell has been extensively adopted due to its compact area and simple structure. Nevertheless, its operation becomes increasingly vulnerable at low supply voltages, particularly during read operations. Read disturb failures, reduced static noise margin (SNM), and increased sensitivity to process, voltage, and temperature (PVT) variations limit the minimum operating voltage of 6T SRAM. These limitations restrict the use of conventional 6T SRAM in energy-constrained applications such as battery-powered systems, wearable devices, and near-threshold computing platforms. To address these challenges, alternative SRAM cell topologies have been explored in recent years, including 7T, 8T, and multi-port SRAM architectures. Among these, the 7T SRAM cell has attracted significant attention due

to its decoupled read path, which isolates the storage nodes from read disturbances. By separating the read and write operations, the 7T SRAM improves read stability and enhances robustness under low-voltage operation. However, this improvement is achieved at the cost of increased transistor count, area overhead, and potential read delay penalties. Therefore, a clear understanding of the performance trade-offs between conventional 6T and 7T SRAM cells is essential for informed memory architecture selection.

Another important aspect of SRAM design is variability tolerance. With continued technology scaling, process variations such as threshold voltage fluctuations and device mismatch significantly impact SRAM yield and reliability. These variations affect key parameters including SNM, access delay, and leakage current, particularly in low-voltage regimes. While several studies report improved stability in 7T SRAM cells, a comprehensive and fair benchmarking against conventional 6T SRAM under identical operating conditions remains necessary to evaluate their true advantages and limitations. In addition, circuit-level simulation and statistical analysis play a crucial role in predicting SRAM behavior under realistic operating conditions. Techniques such as Monte Carlo simulation enable the evaluation of variability-induced performance degradation and provide insights into yield trends. Accurate benchmarking using identical

technology parameters, operating voltages, and simulation setups is essential to ensure meaningful comparison between different SRAM architectures.

II. LITERATURE SURVEY

The development of static random-access memory (SRAM) has played a fundamental role in the evolution of digital integrated circuits, serving as a primary on-chip memory component in processors, caches, and embedded systems. Continuous scaling of CMOS technology has enabled higher integration density, improved speed, and reduced cost; however, it has also introduced significant challenges related to power consumption, variability, and reliability. In particular, SRAM operation at reduced supply voltages has become increasingly difficult due to aggravated read disturb issues, reduced noise margins, and sensitivity to process, voltage, and temperature (PVT) variations. These challenges have motivated extensive research into alternative SRAM cell topologies and design techniques aimed at improving low-voltage robustness.

The conventional 6T SRAM cell has been widely adopted due to its compact area and compatibility with standard CMOS processes. However, several studies have shown that the 6T cell suffers from degraded read static noise margin (RSNM) at low supply voltages because the read operation

directly disturbs the stored data. This limitation restricts the minimum operating voltage and adversely affects yield in deeply scaled technologies. To mitigate these issues, researchers have explored SRAM architectures that decouple the read and write paths, thereby isolating the storage nodes during read operations. Such approaches have been shown to significantly improve read stability and enable reliable operation in near-threshold and subthreshold voltage regimes.

Among the proposed alternatives, 7T and 8T SRAM cells have gained considerable attention. The 7T SRAM cell introduces an additional transistor to separate the read path from the storage nodes, reducing read disturb and improving RSNM. Similarly, 8T SRAM architectures employ fully decoupled read ports to enhance read stability and variation tolerance. While these designs improve low-voltage operation, they often incur penalties in terms of increased area, additional routing complexity, and higher read access delay due to single-ended sensing schemes. Therefore, understanding the trade-offs between stability, performance, and area remains an important research topic.

Several works have also focused on improving SRAM performance through assist techniques such as word-line underdrive, supply voltage boosting, negative bit-line schemes, and adaptive body biasing. These techniques

temporarily modify operating conditions to enhance read or write margins at low voltages. Although effective, assist circuits introduce additional design complexity, power overhead, and sensitivity to timing inaccuracies. As a result, alternative cell-level solutions, such as topology modification and transistor sizing optimization, are often preferred for low-power and variation-aware designs.

Process variability has emerged as a dominant factor affecting SRAM reliability in advanced technology nodes.

Variations in threshold voltage, channel length, and device mismatch significantly impact SNM distributions and access delays,

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